Introduction

Economic freedom denotes an institutional arrangement, in which individuals can exercise personal choice, engage in voluntary exchange, compete in markets, and rely on the protection of property (Gwartney & Lawson, 2003). A stylized fact, in macroeconomics, is its positive association with per capita income (Rondón & Winkelried, 2026). A recent survey reports that the majority of estimates of the economic freedom–growth relationship are positive (Lawson et al., 2024). Most of this literature, however, estimates a coefficient over a prespecified interval. Such estimands do not recover dynamic cumulative responses and, therefore, do not identify when the response emerges, peaks, or dissipates. Notable exceptions are Góes (2015), Rondón and Winkelried (2026), Awan (2025), and Hartley et al. (2026).

Despite recent advances in estimating the dynamic growth effects of economic freedom, a central question remains unaddressed: do economic liberalizations and deteriorations generate symmetric growth responses? Symmetry implies that liberalization retraces the effects of deterioration. By contrast, asymmetry implies that equally sized improvements and deteriorations induce output responses that differ in magnitude. Indeed, Rondón and Winkelried (2026) identify "exploring asymmetries between positive and negative shocks" as a future research direction that "would refine the interpretation of the average response" (p. 12).

To our knowledge, while the theoretical literature provides a rationale for such asymmetries, no cross-country empirical study has examined whether economic freedom liberalizations and deteriorations have asymmetric effects on economic growth. For example, Rodrik (1989) argues that only credible liberalizations induce irreversible investment, whereas deteriorations trigger firm exit and tighter credit constraints. Sunk costs, for their part, imply hysteresis by creating distinct entry and exit thresholds, such that a deterioration may induce firm exit whereas an equally sized improvement may not induce re-entry (Dixit, 1989). Another mechanism arises through trust, as institutional deterioration can shift the economy to a persistent low-entrepreneurship equilibrium that subsequent liberalization may fail to reverse (Aghion et al., 2010).

Quantifying such asymmetries requires impulse responses, the canonical objects for characterizing dynamic effects in macroeconomics (Gonçalves et al., 2024; Kolesár & Plagborg-Møller, 2025). Of particular importance, when shocks are observed and exogenous, impulse responses identify average dynamic causal effects. However, under nonlinear data-generating processes, the estimand "equals a weighted average of the true nonlinear causal effects" (Kolesár & Plagborg-Møller, 2025, pp. 738–739). In other words, the linear estimand still identifies a causal effect, but only as a weighted average of potentially heterogeneous responses. Detecting these nonlinearities is "of first-order concern for policy design" (Caravello & Martínez Bruera, 2024, p. 1). Failure to account for such asymmetries may therefore lead to misleading policy conclusions.

This paper presents the first cross-country dynamic study of these sign asymmetries. In particular, we hypothesize that deteriorations induce greater output losses than liberalizations induce output gains.[1] To operationalize these episodes, we extend the reparameterization of Rondón and Winkelried (2026) to distinguish liberalizations and deteriorations, while accommodating the fact that reforms are enacted, implemented, and recorded in the Economic Freedom of the World

[1] As in all observational research, including instrumental-variables, event-studies, and matching designs, causal interpretation rests on identifying assumptions. We discuss our assumptions in the Methodology section. Robustness analyses assess its sensitivity to alternative controls, measurements, and samples. Neither, however, can verify the restriction. Absent it, the estimates represent conditional dynamic associations.

(EFW) index across adjacent quinquennia. Using a quinquennial panel of up to 164 countries over 1970–2020, we estimate cumulative real GDP-per-capita responses at horizons of 0, 5, 10, and 15 years using local projections (LPs). We further assess the robustness of our findings using alternative shock definitions and specifications.

Consistent with our hypothesis, our estimates reveal substantial asymmetries. At the ten-year horizon, the cumulative GDP-per-capita loss associated with a deterioration is more than four times the cumulative gain associated with a liberalization in absolute log-point magnitude. In general, equality of the sign-specific response paths is rejected jointly across horizons. For policy evaluation, this asymmetry implies that treating deteriorations as the reverse of liberalizations may understate their economic costs. The remainder of the paper is organized as follows. Section 2 reviews the literature. Section 3 presents the methodology. Section 4 presents the empirical results. Section 5 reports the robustness checks. Section 6 concludes.

Literature Review

Institutions comprise the arrangements that structure social interaction, and the incentives facing economic and political actors. In particular, North (1990) defines them as "the rules of the game in a society" (p. 3). Such a social infrastructure generates efficient price signals so that, in the language of North and Thomas (1973), individuals capture the social returns to their actions as private returns (Hall & Jones, 1999). In this regard, Acemoglu et al. (2005) and Acemoglu et al. (2021) argue that economic institutions shape investment, innovation, production, and resource allocation. But, at the same time, these institutions are conceptualized as endogenous collective choices, whose form reflects political power and the distribution of resources. Economic freedom constitutes a component of the institutional domain. While institutions comprise the system of political, social, and economic constraints, economic freedom comprises, in particular, the subset governing property rights, voluntary exchange, open markets, and limits on economic coercion (Doucouliagos & Ulubasoglu, 2006; Gwartney & Lawson, 2003).

As a subset of institutions, economic freedom encompasses the aspect of human freedom related to working, transacting, contracting, owning, and using productive property (Gwartney et al., 2025). It does not, however, encompass civil liberties and political freedom (Dawson, 1998; Doucouliagos & Ulubasoglu, 2006; Fabro & Aixalá, 2012). In the words of Adam Smith, it corresponds to a "system of natural liberty" (Smith, 1776). Yet this emphasis on individual agency should not be equated with laissez-faire (Carlsson & Lundström, 2002; Trantidis & Cowen, 2020). Overall, this conceptualization provides the rationale for its empirical measurement.

Two composite measures prevail in the empirical literature. On the one hand, the Economic Freedom of the World (EFW) index covers up to 165 jurisdictions, from 1970 through 2023, and assigns equal weight to five areas: government size, legal system and property rights, sound money, freedom to trade, and regulation (Gwartney et al., 2025). On the other hand, the Heritage Foundation's Index of Economic Freedom, introduced in 1995, covers up to 184 sovereign countries (Heritage Foundation, 2026). Although Heritage offers more extensive cross-sectional coverage, the longer time dimension of EFW makes it more suitable for studying changes in economic freedom (Gwartney & Lawson, 2003). For that reason, EFW is the most widely used measure in empirical research (Balliew et al., 2020; Bolen & Sobel, 2020; de Haan et al., 2006; Doucouliagos & Ulubasoglu, 2006; Lawson, 2022; Migheli, 2026), thereby facilitating cross-study comparisons

(Caudill et al., 2000; Rondón & Winkelried, 2026; Spindler & Still, 1991).[2]

Nevertheless, the empirical economic freedom–growth relationship, despite relying on the same measure, varies with model specification and time horizon (de Haan et al., 2006; Lawson, 2022; Lawson et al., 2024). de Haan et al. (2006) caution that some studies have serious shortcomings, such as poorly specified growth models. Whether the level of EFW should enter growth regressions, alongside its change, has been the subject of a "vigorous theoretical and empirical debate" (Lawson et al., 2024, p. 1107). Within this debate, for instance, Sturm and de Haan (2001) find that changes in economic freedom exhibit a robust association with growth, whereas its level does not. On balance, the evidence suggests that these specifications identify distinct estimands, each with its own economic interpretation. In our paper, we extend this insight by recognizing that changes may unfold across adjacent periods. We augment the reparameterization of Rondón and Winkelried (2026) to estimate liberalization and deterioration trajectories, recognizing that alternative treatments of persistent shocks recover different dynamic estimands (Alloza et al., 2025).

A relatively understudied aspect of the literature is the temporal evolution of the growth response to EFW changes. Most studies estimate the association over a prespecified interval, without examining when the output response emerges, peaks, or dissipates. Surveys, for instance, organize the EFW–growth evidence around coefficients from regressions, each corresponding to a prespecified growth interval, rather than around responses estimated separately at each horizon (de Haan et al., 2006; Lawson et al., 2024). Recent exceptions adopt distinct approaches to estimating dynamic responses. For example, Awan (2025) uses a Bayesian panel VAR for low- and lower-middle-income countries, Rondón and Winkelried (2026) apply panel LPs to quinquennial changes in log EFW over 1970–2020, and Hartley et al. (2026) trace responses around large, discrete reform episodes. Of particular importance, Rode and Coll (2012) and Rondón and Winkelried (2026) argue that the estimated effects of economic freedom are horizon-dependent, underscoring the need to estimate responses separately at each horizon.

Several mechanisms may account for the observed variation in the growth response across horizons. Dawson (1998) identifies two channels through which economic freedom affects growth: a direct productivity channel and an indirect channel operating through physical-capital accumulation. Consistent with this, Doucouliagos and Ulubasoglu (2006) and Rondón and Winkelried (2026) report attenuated estimates after controlling for physical capital. Doucouliagos and Ulubasoglu (2006) also find lower estimates in panel models, which they interpret as evidence that transitory effects are less pronounced than long-run effects. Another perspective is provided by Bolen and Williamson (2019), who show that institutional trajectories contain information beyond levels and changes, and that greater EFW volatility is associated with lower growth. Overall, these mechanisms imply a horizon-dependent output response, as capital accumulation, investment, and productivity under uncertainty adjust at different rates.

Turning to identification strategies, most studies employ a cross-sectional design, in which period averages or initial levels of economic freedom are regressed on period-average economic growth (Justesen, 2008). Using panel data and Granger-causality tests, Dawson (2003), Heckelman (2000) and Vega-Gordillo and Álvarez-Arce (2003) find that economic freedom precedes growth. However, Granger-causality tests establish predictive ordering rather than structural causation. For their part, instrumental-variable estimates remain assumption-dependent, because

[2]As with any composite indicator, the EFW index is sensitive to measurement and aggregation choices (Balliew et al., 2020; Bolen & Sobel, 2020; Carlsson & Lundström, 2002; Caudill et al., 2000). We therefore examine component-specific responses and leave-one-component-out EFW indices.

they rely on the validity of the exclusion restriction. For instance, Faria and Montesinos (2009) obtain positive results using latitude, legal origin, ethnolinguistic fractionalization, and settler mortality as instruments, whereas Jordaan (2023) reports similar results from an instrumental-variable three-stage system. As alternative identification approaches, Grier and Grier (2021) match large, sustained EFW increases to similar untreated cases, and Hartley et al. (2026) use a staggered event-study design to estimate dynamic effects around major reform episodes. Across these approaches, matching relies on conditional independence and overlap, and event-study designs on parallel trends and no anticipation.

To the best of our knowledge, despite recent advances in estimating dynamic responses, the literature has yet to examine whether increases and declines in economic freedom generate symmetric growth trajectories. Nevertheless, the theoretical literature provides reasons for expecting such asymmetries. Beyond those discussed in the introduction, on the investment side, Bernanke (1983) describes investment under irreversible capital and information arriving over time as "a problem in stochastic dynamic optimization" (p. 87). For that reason, in each period, the potential investor must decide whether to commit himself immediately or to defer commitment (Bernanke, 1983). This implies that reforms may have delayed effects as investors wait for adequate institutional conditions, whereas reversals may discourage investment immediately. A complementary political-economy mechanism reinforces the case for asymmetric growth responses. Fernandez and Rodrik (1991) show that if there is a chance to reconsider a reform, it may be repealed because adoption reveals how individuals actually fare under the reform. By contrast, when the reform is rejected, "no new information is revealed, since the status quo is maintained" (p. 1149).

Viewed as a whole, the literature shows that the positive relationship between economic freedom and growth varies across horizons and that evaluating reforms, therefore, requires dynamic estimands. Yet whether liberalizations and deteriorations generate symmetric output trajectories remains unknown. Theoretical arguments, for their part, provide reasons to expect that deteriorations generate larger output losses than the gains associated with equally sized liberalizations. If this asymmetry holds systematically across countries, the symmetric estimates that dominate the literature conflate distinct sign-specific response paths and, therefore, do not recover the relevant dynamic effects, fundamentally altering the interpretation of the existing evidence.

Methodology

Estimating dynamic responses relies on either VARs or LPs. With unrestricted lag structures, both methods recover the same population impulse responses (Plagborg-Møller & Wolf, 2021). However, in finite samples, VARs are more sensitive to misspecification, because, when iterated, "misspecification errors are compounded with the forecast horizon" (Jordà, 2005, p. 162). In constrast, LPs separately estimate each horizon and impose fewer cross-horizon restrictions, reducing sensitivity to dynamic misspecification bias (Jordà, 2005; Montiel Olea et al., 2025). Moreover, estimating each horizon with LPs also facilitates nonlinear specifications (Jordà, 2005; Jordà, 2023). Recent studies formalize the identification and interpretation of LPs under sign and size nonlinearities (Caravello & Martínez Bruera, 2024) and state dependence (Cloyne et al., 2023; Gonçalves et al., 2024). These properties make LPs particularly advantageous for estimating sign-specific response paths, while allowing dynamic responses to vary flexibly across horizons.

We express real GDP per capita in logarithms to interpret changes as approximate percentage

changes. In practice, levels and long-differenced LPs target the same population response, but the latter can improve finite-sample performance. For externally identified observed shocks, it "substantially reduces, and in many cases eliminates, estimation bias, as well as significantly improves confidence interval coverage" (Piger & Stockwell, 2025, p. 759).

Moreover, in this paper, we use a quinquennial frequency. Dawson (1998) argues that "a five-year subperiod is probably the shortest term in which it can be reasonably expected that all business cycle fluctuations will be 'averaged' out" (p. 607). Quinquennial spacing also mitigates measurement noise and recording delays inherent in slow-moving institutional change (Bolen & Sobel, 2020; Dawson, 1998, 2003; Fabro & Aixalá, 2012; Farhadi et al., 2015; Rondón & Winkelried, 2026; Vega-Gordillo & Álvarez-Arce, 2003).

Outcome and shock. Let $y_{i,t} = \log(Y_{i,t})$ denote log real GDP per capita in country i at period t, and let $s_{i,t}$ denote a change in economic freedom. At horizon h, we define the outcome as the cumulative change, from the pre-change date $t-1$ to the endpoint $t+h$, for $h = 0, 1, 2, 3$. The four endpoints are, therefore, 0, 5, 10, and 15 years, after period t. Our long-difference formulation directly targets a cumulative response.

Sign asymmetry. To allow liberalizations and deteriorations to have different response slopes, we split the change at zero, as in Barnichon et al. (2022),

$$s^+_{i,t} = \max(s_{i,t}, 0), \qquad s^-_{i,t} = \min(s_{i,t}, 0).$$

This is algebraically equivalent to including the signed change and its absolute value. The coefficients are sign-specific average projection slopes over the observed support (Caravello & Martínez Bruera, 2024; Kolesár & Plagborg-Møller, 2025).

Panel local projections and reparametrization. We estimate a separate panel regression for each horizon $h \in \{0, 1, 2, 3\}$. We define the first differences of the sign-specific shocks as $\Delta s^+_{i,t} = s^+_{i,t} - s^+_{i,t-1}$ and $\Delta s^-_{i,t} = s^-_{i,t} - s^-_{i,t-1}$. The asymmetric panel LP is,

$$\begin{aligned} y_{i,t+h} - y_{i,t-1} = \alpha_i^{(h)} + \lambda_t^{(h)} + \beta_h^+ s^+_{i,t} + \beta_h^- s^-_{i,t} \\ + \phi_h^+ \Delta s^+_{i,t} + \phi_h^- \Delta s^-_{i,t} + \rho_h \Delta y_{i,t-1} \\ + \boldsymbol{\gamma}_h' \mathbf{X}_{i,t-1} + \varepsilon_{i,t}^{(h)}, \qquad h = 0, 1, 2, 3. \end{aligned}$$

The two first differences of the sign-specific shocks arise from the same exact reparametrization of a one-lag distributed-shock model. Using the superscript $\pm$ to denote either sign,

$$\begin{aligned} \delta^\pm_{0,h} s^\pm_{i,t} + \delta^\pm_{1,h} s^\pm_{i,t-1} &= \beta^\pm_h s^\pm_{i,t} + \phi^\pm_h \Delta s^\pm_{i,t}, \\ \beta^\pm_h &= \delta^\pm_{0,h} + \delta^\pm_{1,h}, \qquad \phi^\pm_h = -\delta^\pm_{1,h}, \\ \delta^\pm_{0,h} &= \beta^\pm_h + \phi^\pm_h. \end{aligned}$$

This reparameterization is proposed by Rondón and Winkelried (2026). In particular, we expand it for our asymmetric setting. As persistence is part of the economic object, reforms may unfold beyond one quinquennium, even as five-year averaging attenuates business-cycle noise (Dawson, 1998). Thus, our treatment is a signed reform path, in the spirit of the assignment-path vector of Rambachan and Shephard (2025) and the "treatment plan" of Dube et al. (2025). This logic also underlies the sustained-jump definition of Grier and Grier (2021) and the major-episode design of Hartley et al. (2026). As a final consideration, Rondón and Winkelried (2026) further motivate this approach by noting that reforms unfold through a sequence of announcement, legal

adoption, administrative implementation, enforcement, and private adjustment.

Our main estimates are the coefficients ($\beta_h^{\pm} = \delta_{0,h}^{\pm} + \delta_{1,h}^{\pm}$) for liberalizations and deteriorations. They correspond to a one-point increase or decrease in the EFW index in each of two adjacent quinquennia, depending on the sign of the reform. We consider $\beta_h^{\pm}$ because they permit a direct test of whether equal-sized, equal-duration liberalizations and deteriorations generate symmetric output responses. This combined-path estimand better captures the inherently persistent nature of reforms, which typically unfold across multiple periods. Indeed, the 77 major two-point EFW liberalizations studied by Lawson and Lawson (2020) last approximately 15 years on average, with durations ranging from 5 to 39 years.

Moreover, including both adjacent changes prevents the current coefficient from absorbing the lagged response, when reforms are persistent. Our specification allows an episode recorded across adjacent observations to enter through both periods. In addition, conditioning on $\Delta y_{i,t-1}$ compares countries with comparable recent output dynamics. Prior growth is measured over the same quinquennium as $s_{i,t-1}$. Thus, it may already reflect part of the response to the preceding EFW change. Holding it fixed, therefore, conditions out adjustment realized before $t-1$. The same consideration applies to other $t-1$ controls that may have responded to $s_{i,t-1}$. Hence, $\beta_h^{\pm}$ identifies a conditional continuation response from $t-1$ onward, rather than the unrestricted total effect of a reform path beginning at $t-2$. We report our two main estimates in our Results section, with their interpretations. First, we present the main "path" estimates β_h^{+} and β_h^{-}. Second, we report the current-change estimates ($\delta_{0,h}^{\pm} = \beta_h^{\pm} + \phi_h^{\pm}$). These estimates are reported alongside their statistical significance and other relevant summary measures.

Lag augmentation and fixed effects. We lag-augment each horizon regression with prior five-year GDP growth. In this sense, Montiel Olea and Plagborg-Møller (2021) show that, under their regularity conditions, lag augmentation obviates the need for an additional HAC correction for serial correlation in the residuals. Meanwhile, country fixed effects $\alpha_i^{(h)}$ absorb time-invariant heterogeneity and time fixed effects $\lambda_t^{(h)}$ absorb global conditions common to each quinquennium.

Identification. A causal interpretation of $\delta_{0,h}^{\pm}$ requires no anticipation or cross-country interference, conditional exchangeability with potential GDP paths, a correctly specified adjustment and response function, and conditional positivity. Interpreting $\beta_h^{\pm}$ as a causal response additionally requires sequential exchangeability and joint positivity at both treatment dates, an additive path-response function, and no relevant treatment-induced confounding (Angrist & Kuersteiner, 2011; Kolesár & Plagborg-Møller, 2025; Rambachan & Shephard, 2025).

Several features of EFW movements, together with the diagnostic tests and robustness exercises, lend plausibility to the identifying assumptions, but cannot verify them. Thus, we distinguish between two interpretations of the estimates. Under the stated identifying assumptions, they admit a causal interpretation as the effects of sign-specific EFW paths. Absent these assumptions, they should be interpreted as conditional dynamic associations.

Previous studies provide arguments for treating certain EFW movements as plausibly exogenous, and propose approaches to improve identification. For example, Rondón and Winkelried (2026, p. 7) characterize the major Peruvian reform episodes as "discrete, politically driven, and difficult to interpret as endogenous responses to short-run fluctuations in output" and relate them to the Latin American sequence of nationalization, deterioration, and liberalization. Furthermore, Hartley et al. (2026) find no differential pre-trends, when using an indicator for major EFW reform episodes. This finding suggests that major reforms are not preceded by growth accelerations.

However, discrete event-study treatment differs from our continuous, sign-specific specification, so their evidence provides only complementary support.

To strengthen the identification of reform episodes, Lawson and Lawson (2020) use the dispersion of changes across the five EFW areas to distinguish broad-based reforms from movements concentrated in only one or two components. In the same spirit, Rondón and Winkelried (2026) use percentile- and rank-based shock definitions to identify major reform episodes. In general, the stability of the results across exercises supports the identification strategy by reducing the plausibility that the findings are artifacts of anticipation, omitted variables, measurement choices, or influential observations.

Inference. Outcomes associated with adjacent projection dates cover overlapping years and, therefore, share some of the same GDP innovations. This overlap makes the regression errors serially correlated, while their variances may also differ across countries and dates. To accommodate these features, for panel inference, we use cluster standard errors by country. It allows for heteroskedasticity and serial correlation within each country.

These adjustments provide valid inference for each horizon, but they do not yield a confidence region for the response path as a whole. Pointwise intervals are not confidence regions for the entire response path, because the horizon-specific coefficients are correlated (Inoue et al., 2026; Jordà, 2009). Valid pathwise inference must, therefore, account jointly for the family of hypotheses across horizons. For that reason, in addition to pointwise intervals, we construct 95-percent simultaneous *sup-t* bands, and also conduct joint Wald tests of zero response and sign symmetry (Inoue et al., 2026).

To be implemented, we follow the plug-in *sup-t* procedure of Montiel Olea and Plagborg-Møller (2019). We consider an estimated response vector $\widehat{\boldsymbol{\beta}} = (\widehat{\beta}_h)_{h\in\mathcal{H}}$, where $\mathcal{H} = \{0, 1, 2, 3\}$. Let $\widehat{\boldsymbol{\Sigma}}$ denote the estimated cross-horizon covariance matrix of $\widehat{\boldsymbol{\beta}}$, and define $\widehat{\mathbf{D}} = \text{diag}\{\widehat{\sigma}_h : h \in \mathcal{H}\}$. The estimated cross-horizon correlation matrix and the simultaneous critical value are,

$$\widehat{\mathbf{R}} = \widehat{\mathbf{D}}^{-1}\widehat{\boldsymbol{\Sigma}}\widehat{\mathbf{D}}^{-1}, \qquad c_{0.95} = Q_{0.95}\left(\max_{h\in\mathcal{H}}|Z_h^*|\,\middle|\,\widehat{\mathbf{R}}\right), \qquad \mathbf{Z}^* \sim N(\mathbf{0}, \widehat{\mathbf{R}}),$$

where $Q_{0.95}$ denotes the 95th percentile of the simulated maximum absolute standardized statistic, conditional on $\widehat{\mathbf{R}}$. Then, the simultaneous interval at horizon h is,

$$\left[\widehat{\beta}_h - c_{0.95}\widehat{\sigma}_h,\ \widehat{\beta}_h + c_{0.95}\widehat{\sigma}_h\right], \qquad h \in \mathcal{H}.$$

The critical value incorporates the estimated dependence across horizons. Thus, the intervals jointly cover the response vector at all horizons in $\mathcal{H}$ with 95 percent probability asymptotically.

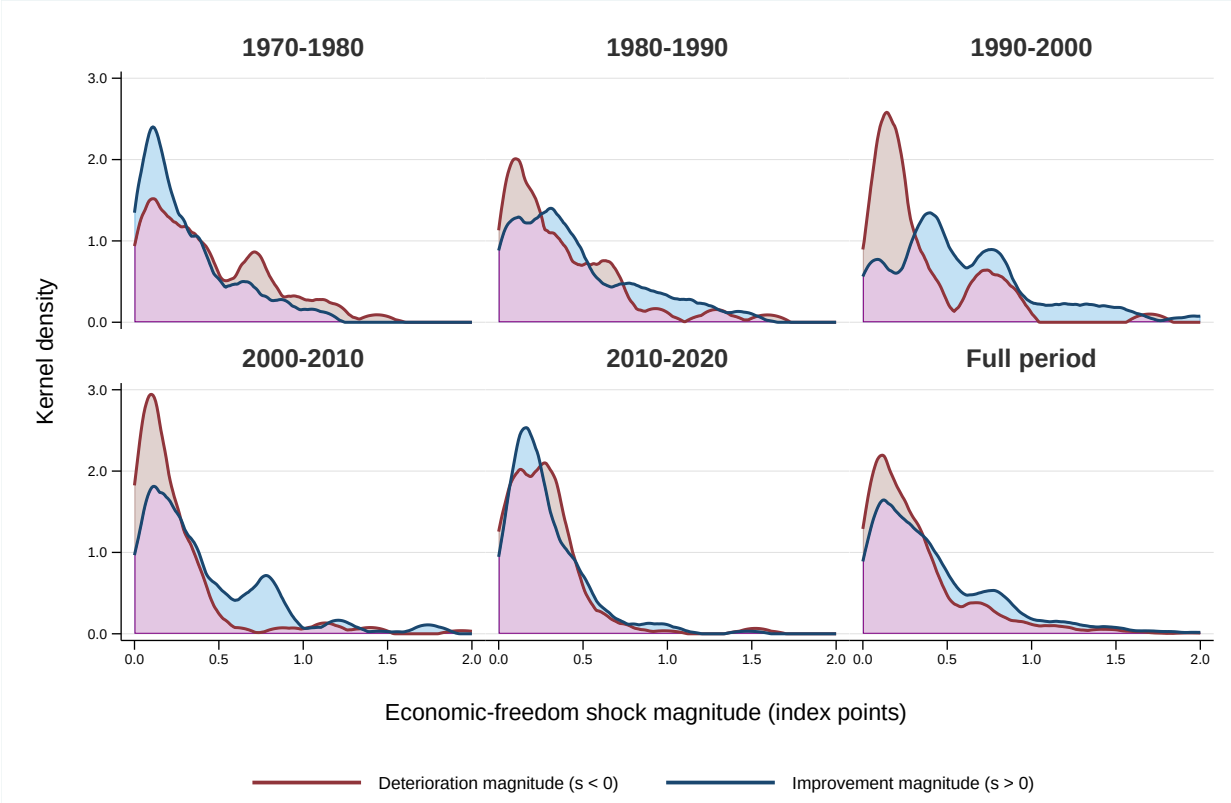
Data

For this paper, we construct a country-level panel observed at five-year intervals from 1970 to 2020. The resulting sample contains 1,451 country–year observations covering 164 countries, and integrates five data sources. First, the World Development Indicators provide real GDP per capita and population (World Bank, 2026). Second, the Fraser Institute supplies the EFW score and its five component areas (Gwartney et al., 2025). Third, V-Dem provides the Electoral Democracy Index (Coppedge et al., 2026). Fourth, the United Nations National Accounts Main Aggregates database supplies sectoral value added and expenditure aggregates (United Nations Statistics Divi-

sion, 2026). Fifth, Penn World Table 11.0 supplies total factor productivity, capital, employment, and human capital (Feenstra et al., 2015, 2025). We retain missing values as missing.

Our main outcome uses World Bank real GDP per capita in constant 2015 U.S. dollars. Appendix Figure A1 illustrates cross-country heterogeneity in GDP per capita. The augmented Solow model of Mankiw et al. (1992) attributes these differences proximately to variation in physical- and human-capital accumulation and population growth. Subsequent research identifies economic freedom, as an institutional determinant of growth through productivity and investment (Dawson, 1998; Hall & Jones, 1999). On the other hand, our variable of interest is the five-year change in the EFW index, separated by sign to capture asymmetries. Appendix Figure A2 documents the evolution of five-year EFW changes. The distribution shifts most strongly toward improvements during the 1990s. In 1995, the median reaches 0.41 index points and the 95th percentile reaches 1.62 points. Yet improvements and deteriorations coexist in every quinquennium.

Figure 1: Distributions of improvement- and deterioration-shock magnitudes, by decade.



Notes: Each panel compares kernel densities of the shock magnitude $|s_{i,t}|$ for deteriorations ($s_{i,t} < 0$, red) and improvements ($s_{i,t} > 0$, blue), where $s_{i,t} = EFW_{i,t} - EFW_{i,t-1}$. Purple shows their overlap. The five decade panels group shocks by endpoint year: $1970 < t \leq 1980$, $1980 < t \leq 1990$, $1990 < t \leq 2000$, $2000 < t \leq 2010$, and $2010 < t \leq 2020$. The last panel pools all 1,281 nonzero shocks: 515 deteriorations and 766 improvements. Densities are normalized within sign and panel and use a common bandwidth, so height shows concentration rather than observation counts.

Figure 1 further illustrates the historical evolution of shock magnitudes across decades. Improvement shocks shift away from zero through the 1980s and reach their greatest separation during 1990–2000, consistent with Grier and Grier (2021), who interpret these episodes as generalized economic reforms reflecting the main tenets of the Washington Consensus. The timing also

aligns with post-socialist transition. Lawson and Lawson (2020, p. 674) observe that "most of the formerly communist nations reformed quickly in the early- and mid-1990s." These episodes form part of the cycle identified by Alesina et al. (2020), in which liberalization spread across advanced and developing economies from the late 1980s, but slowed after the Global Financial Crisis. This decline is reflected in the subsequent near-overlap of the densities during 2010–2020.

Table 1: Descriptive Statistics

	N	Mean	SD	P25	Median	P75	Min	Max
Real GDP per capita	1,451	11,699	16,622	1,360	4,103	16,146	139	117,073
Five-year GDP per capita growth	1,287	0.082	0.160	0.003	0.087	0.165	-1.184	0.779
EFW summary score	1,451	6.141	1.296	5.232	6.162	7.112	2.221	9.199
Five-year EFW shock	1,287	0.128	0.503	-0.150	0.087	0.380	-1.938	2.845
Positive-shock component	1,287	0.252	0.360	0.000	0.087	0.380	0.000	2.845
Negative-shock component	1,287	-0.124	0.248	-0.150	0.000	0.000	-1.938	0.000
Lagged five-year GDP per capita growth	1,126	0.093	0.157	0.017	0.095	0.171	-0.588	0.779
Lagged electoral democracy	1,272	0.491	0.283	0.217	0.480	0.775	0.013	0.921
Lagged agriculture share	1,285	0.142	0.144	0.034	0.087	0.203	0.000	0.687
Lagged manufacturing share	1,280	0.144	0.073	0.094	0.139	0.181	0.003	0.600
Lagged population growth	1,126	0.085	0.073	0.032	0.081	0.131	-0.262	0.672

Notes: Statistics use the full available quinquennial panel. Each row includes every nonmissing observation for that variable, so sample sizes differ. EFW is measured on a zero-to-ten scale, and shocks are index-point changes. Sectoral variables are shares. The positive component is nonnegative and the negative component is nonpositive.

Table 1 show that real GDP per capita is right-skewed. Its mean is $11,699, but the median is only $4,103, showing that a small number of rich countries raise the average. The median five-year GDP change is approximately 9 percent. The mean EFW score is approximately 6.1, and its median five-year change is 0.087 index points. Nevertheless, the extremes reveal occasional reforms and reversals of much greater magnitude. The variation in democracy, sectoral composition, and population growth shows that the panel covers countries with very different political and economic structures. Together, these contrasts provide the variation needed to examine how GDP evolves after improvements and deteriorations in economic freedom.

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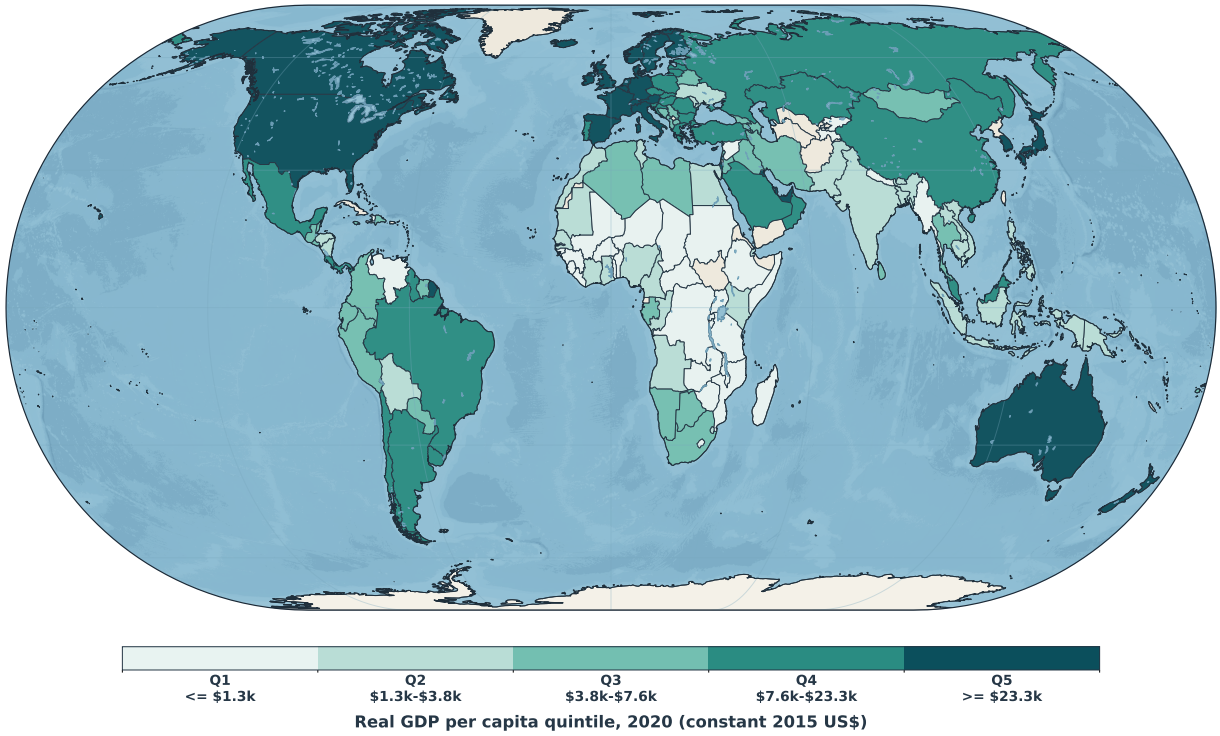
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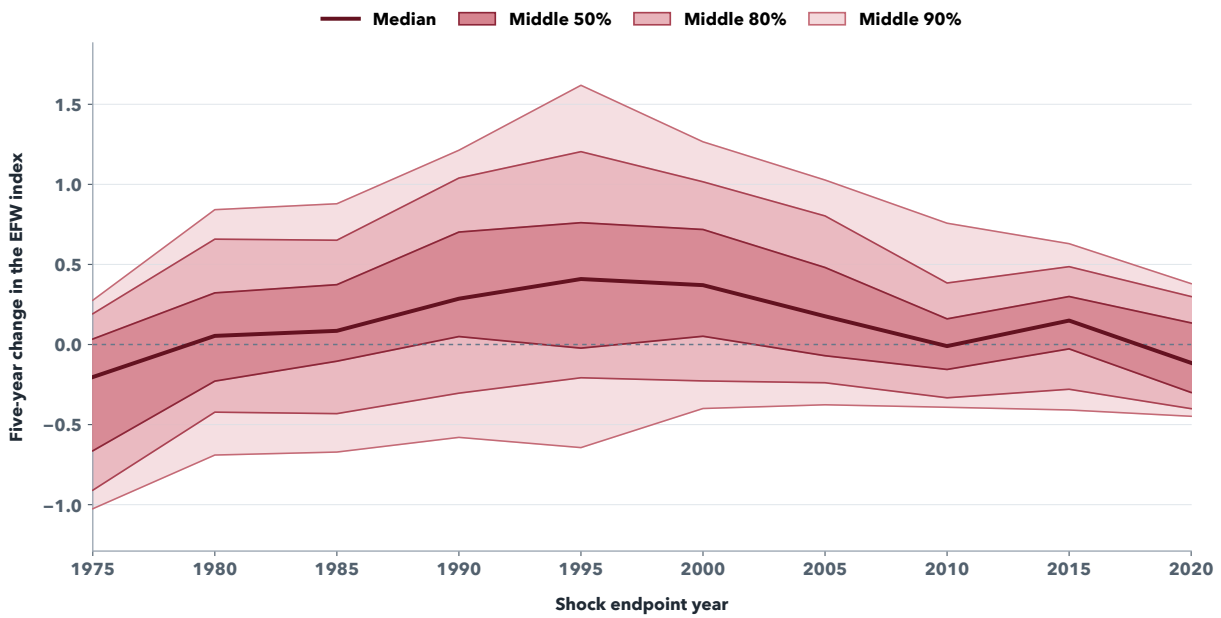
Appendix

Figure A1: World distribution of real GDP per capita in 2020, by quintile.



Notes: Countries are assigned to quintiles of the cross-country distribution of real GDP per capita in 2020, measured in constant 2015 U.S. dollars. Q1 is the lowest-income quintile and Q5 is the highest-income quintile. The legend reports the approximate quintile cutoffs. The map covers the 163 countries with a positive, nonmissing GDP-per-capita observation in 2020. Countries without a valid observation are shown in beige and are not assigned to a quintile. Source: World Development Indicators.

Figure A2: Distribution of five-year changes in economic freedom, 1975–2020.



Notes: The figure summarizes the cross-country distribution of $s_{i,t} = EFW_{i,t} - EFW_{i,t-1}$ at each quinquennial endpoint in the Experiment 3 chain-linked panel. The dark line is the median; the nested shaded regions contain the middle 50 percent (p25–p75), 80 percent (p10–p90), and 90 percent (p5–p95) of shocks. A shock is observed only when a country has nonmissing real GDP per capita and EFW at t and an adjacent retained observation at $t-1$. The sample contains 1,287 five-year changes across 161 countries. The dashed zero line separates improvements from deteriorations. The figure is descriptive and does not identify causal effects. Source: Fraser Institute's EFW Panel Dataset and World Development Indicators.